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REVIEW

Eye tracking as a digital biomarker in neurodegenerative diseases

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Abstract

Oculomotor abnormalities are a common finding in neurodegenerative diseases due to degeneration of neural pathways and brain regions involved in controlling eye movements. Pathological changes to the dorsolateral prefrontal cortex, basal ganglia, superior colliculus and cerebellum produce subtle changes in eye-movement metrics that may not be detected by clinical examination. The present review addresses the potential use of eye-movement biomarkers in neurodegenerative conditions such as multiple sclerosis, Parkinson's disease, Alzheimer's disease and other dementias, and amyotrophic lateral sclerosis. Eye-movement metrics such as saccades, anti-saccades, fixation and smooth pursuit are prognostic of disease progression, can differentiate pathologic subtypes as an aid to diagnosis, and enable clinicians to evaluate early worsening of motor and cognitive function. The cost of medical technologies limits their optimal use and accessibility in clinical practice. The shortage of subspecialist neurologists further limits access to care. New eye-tracking technologies incorporated into widely-accessible digital devices such as smart phones and tablets now permit detailed assessments with minimal equipment requirements, providing an important non-invasive and potentially cost-effective method for patient evaluation in routine clinical practice and as an aid to treatment decision-making. Digital biomarkers can be readily employed by healthcare professionals such as family physicians, nurses and pharmacist to bridge the care gaps, potentially providing them with powerful tools that can be broadly adopted to improve the delivery of care to patients with neurodegenerative conditions.

Keywords Eye movements · Oculomotor · Digital biomarkers · Neurodegenerative diseases · Multiple sclerosis · Parkinson's disease · Alzheimer's disease

Introduction

Oculomotor abnormalities occur across a range of neurological diseases due to degeneration of neural pathways and brain regions involved in controlling eye movements. Subtle changes in eye-movement metrics such as saccades, anti-saccades, fixation and smooth pursuit are prognostic of disease progression and enable clinicians to evaluate early worsening of motor and cognitive function. Patterns of oculomotor abnormalities have the potential to aid in screening and diagnosis and to detect early physical and cognitive deficits.

The assessment of eye movements entered clinical practice in 1962 with the advent of electro-oculography, which measured corneo-retinal electrical potentials. Although initially developed to evaluate retinal function [1], it soon became a valuable tool for quantifying eye movements. Infrared television-based systems and video-oculographic recordings were less invasive and offered greater precision in evaluating eye movements. Recent improvements in eye-tracking technologies powered by artificial intelligence (AI) now permit eye-movement biomarkers (EMB) to be obtained with minimal specialized equipment requirements, providing an important non-invasive method for patient evaluation. These technologies are becoming essential in neurological

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practice as the advent of novel therapeutics and more intensive investigations have heightened the need for accessible, cost-effective tools to evaluate and monitor patients in a more rigorous and timely manner.

The present review was conducted as a selective narrative synthesis of the literature on eye-movement biomarkers in neurodegenerative diseases. We searched PubMed and Google Scholar for articles published between 2000 and 2025 using combinations of terms such as eye tracking, oculomotor, saccades, smooth pursuit, fixation, biomarkers, multiple sclerosis, Parkinson's disease, Alzheimer's disease, dementia and amyotrophic lateral sclerosis. Additional studies were identified from reference lists of relevant reviews and primary articles. We prioritized peer-reviewed human studies reporting quantitative eye-movement metrics and excluded case reports, technical demonstration papers without clinical relevance, and non-English publications. Because this was not a systematic review, formal study-quality scoring was not performed; however, we incorporated qualitative assessment of methodological strengths and limitations in the disease-specific syntheses.

The review addresses the potential use of EMB in neurodegenerative conditions such as multiple sclerosis, Parkinson's disease, Alzheimer's disease and other dementias, and amyotrophic lateral sclerosis. The term EMB is used as an umbrella concept that includes both task-based oculomotor metrics (e.g. saccadic latency, anti-saccade error rate, pursuit gain) and gaze-pattern measures obtained during free-viewing or visual-exploration paradigms. It should be noted that there are also practical applications for EMB in other neurocognitive and neuropsychiatric conditions as well (Table 1).

Neuropathological substrate of eye movement disorders

Oculomotor abnormalities may arise due to neurodegenerative changes or lesions in the neural circuits and brain regions that govern eye movements. The brainstem circuitry controls reflexive, automatic eye movements whereas the oculomotor cortical circuitry controls conscious eye movements initiated by perception and intention. The most commonly evaluated eye movements are smooth pursuit, which is the ability to visually track a moving object; fixation, which maintains steady gaze on a stationary target; and saccades, which are rapid eye movements that reorient gaze toward new points of interest. Saccades may be voluntary or memory-guided (to a previously remembered object). Anti-saccades by contrast assess inhibitory control by requiring suppression of the reflexive urge to look toward a visual stimulus and the voluntary generation of a saccade in the opposite direction.

Voluntary saccades are largely initiated in the frontal, supplementary and parietal eye fields in the cortex (Fig. 1) [37]. The frontal and supplemental eye fields predominate in the generation of volitional or cognitively demanding saccades whereas the parietal eye fields are more important in generating reflexive saccades [38, 39]. Cortical signalling converges in the superior colliculus in the midbrain, which outputs to the paramedian pontine reticular formation (PPRF) in the brainstem (horizontal saccades), and the rostral interstitial nucleus of the medial longitudinal fasciculus (MLF) in the midbrain (vertical saccades) [40]. The brainstem circuits output to the third, fourth and sixth cranial nerve nuclei (oculomotor, trochlear and abducens) that control the oculomotor muscles. Excitatory and inhibitory burst neurons regulate the direction of the saccade and are silent during fixation.

This primary saccadic pathway is modulated by the cerebellum and basal ganglia. Cerebellar projections to the excitatory and inhibitory burst neurons modulate the duration and amplitude of saccades, accelerating or decelerating the saccade to improve accuracy [41]. Further saccade modulation is performed by the basal ganglia circuit. During a voluntary saccade, cortical input activates the caudate nucleus, which in turn reduces the GABAergic inhibitory output from the substantia nigra pars reticulata to the superior colliculus, thereby permitting initiation of voluntary saccades [42].

Smooth pursuit eye movements are used to track objects in motion. Movement is detected by the medial temporal cortex and the frontal eye fields, which initiate signalling to the dorsolateral nuclei of the pons and the vestibulocerebellum. Information is then relayed to the ocular motor nuclei via the vestibular nuclei. Gaze fixation is regulated by the rostral superior colliculus. Signalling from the retina and occipital lobes inhibits neurons in the caudate superior colliculus and dorsal raphe, which inhibits saccades and allows for gaze fixation. Fixation stability relies on cortical and subcortical control systems, cerebellar modulation, and the activity of brainstem oculomotor integrator networks that support gaze-holding and suppress intrusive saccades. Disruption within these circuits can contribute to abnormalities in fixation and microsaccadic behaviour.

EMB enable the identification of neurodegenerative changes in brain regions or neural circuits that regulate eye movements [12]. For example, the most common eye-movement disorder in early Parkinson's disease is a reduced amplitude (hypometria) of volitional saccades due to neurodegenerative changes to substantia nigra pars reticulata projections to the superior colliculus. Other eye-movement abnormalities, such as increased saccadic latency (time to initiate saccades), occur later in the disease course, presumably due to more widespread damage to non-dopaminergic circuits [43].

Table 1 Neurological disorders with distinct oculomotor abnormalities

Disorder	Primary oculomotor abnormalities	Key diagnostic/hallmark features
Alzheimer's disease (AD) [2–7]	Fixation: ↑ instability, ↑ saccadic intrusions Reflexive Saccades: ↓ velocity, hypometric, ↑ latency Anti-saccades: ↑ error rates, ↑ latency	Anti-saccade error rate (potential early biomarker – may precede memory symptoms)
Frontotemporal dementia (Ftd) [8–10]	Fixation: ↑ instability, ↑ saccadic intrusions Reflexive Saccades: relatively preserved Anti-saccades: ↑ error rates	Anti-saccade error rate (earlier onset and more severe than in AD); preserved reflexive saccades contrasts with AD
Dementia with lewy bodies (DLB) [11–13]	Fixation: ↑ instability, ↑ saccadic intrusions Reflexive Saccades: ↓ velocity (vertical), hypometric, ↑ latency Anti-saccades: ↑ errors, ↑ latency	Prolonged saccadic latency (in both saccade and anti-saccade – can help distinguish DLB from AD)
Parkinson's disease (PD) [14–17]	Fixation: ↑ instability, ↑ saccadic intrusions, ↑ SWJ Reflexive Saccades: hypometric, stepwise Anti-saccades: ↑ errors, ↑ latency (mostly in advanced stages)	Hypometric, stepwise saccades are characteristic
Progressive supranuclear palsy (PSP) [18, 19]	Fixation: ↑ instability, ↑ saccadic intrusions, ↑ SWJ Reflexive Saccades: hypometric (vertical > horizontal); ↓ velocity (vertical > horizontal) Anti-saccades: ↑ errors, ↑ latency	Profound slowing of vertical saccades (especially downward) with restricted range distinguishes PSP from PD and AD; increased SWJ frequency and increased anti-saccade impairment can also help differentiate from PD
Multiple system atrophy (MSA) [20–22]	Fixation: ↑ instability, ↑ saccadic intrusions, ↑ SWJ Reflexive Saccades: ↓ velocity, some evidence of hypometria/stepwise Anti-saccades: ↑ errors, ↑ latency	SWJ: PD < MSA < PSP Saccadic slowing resembles PSP but without vertical-specific palsy; slowing helps distinguish from PD
Corticobasal degeneration (CBD) [23–25]	Fixation: possible mild instability Reflexive Saccades: hypometric, ↑ latency + impaired initiation Anti-saccades: ↑ errors, ↑ latency	Asymmetric ocular apraxia (saccade initiation failure), unlike the symmetric vertical slowing of PSP
Huntington's disease (HD) [26–28]	Fixation: ↑ instability, ↑ saccadic intrusions Reflexive Saccades: relatively preserved Anti-saccades: ↑ errors, ↑ latency	Marked voluntary saccade initiation and antisaccade impairment, with relative preservation of reflexive saccades
Multiple sclerosis (MS) [29–32]	Fixation: ↑ instability (often nystagmus-related) Reflexive Saccades: ↑ latency, ↓ velocity Anti-saccades: ↑ errors, ↑ latency	Internuclear ophthalmoplegia (INO) — asymmetric adduction impairment with contralateral abducting nystagmus and saccadic slowing
Amyotrophic lateral sclerosis (ALS, ALS-FTD spectrum) [33–36]	Fixation: Generally stable Reflexive Saccades: largely preserved in pure ALS; subtle ↑ latency or hypometria may appear with frontotemporal involvement (ALS-FTD) Anti-saccades: ↑ errors, ↑ latency, most pronounced in ALS-FTD	Oculomotor function is preserved in classic ALS; antisaccade impairment signals frontotemporal involvement and parallels the ALS-FTD spectrum

SWJ square-wave jerks

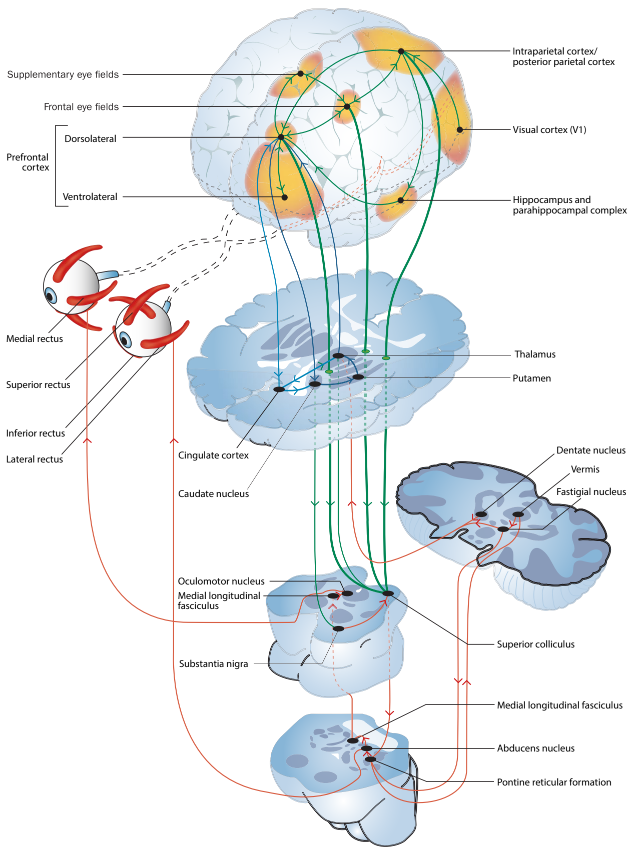

Fig. 1 Neuronal networks involved in saccades and anti-saccades. Executive (light blue) and memory (dark blue) circuits maintain attention toward or away from a visual target and relay signals via oculomotor pathways (green lines) to the superior colliculus, which in turn drives saccades through brainstem motor circuits. Adapted from Fielding et al. [37] Permission for reuse has been confirmed and documented with the publisher

Similarly, in patients with multiple sclerosis (MS), worsening saccadic metrics have been shown to be associated with more extensive demyelination and more severe cortical grey-matter atrophy, which are associated with more severe physical and cognitive impairment [44].

Eye-movement testing

Eye-movement evaluation is an essential component of the neurological examination of patients with neurodegenerative diseases. The clinical examination evaluates eye fixation stability (the nine primary, secondary and tertiary positions of gaze), smooth pursuit with the H test, horizontal and vertical saccades, and convergence with the Near Point of Convergence test, which can detect nystagmus or saccadic intrusions [45]. The limitation of clinical examination, however, is that findings are qualitative and subjective, which may result in underdetection of abnormalities, especially subtle changes related to evolving pathology. For example, in a study of internuclear ophthalmoparesis (INO) in MS patients with mild adduction slowing, INO was not clinically identified by 71% of physician observers [46]. Thus, there is a need for more sensitive quantitative testing.

The most common oculomotor tasks that are performed in eye-movement evaluations include (Fig. 2):

Saccade task, in which subjects must reorient gaze toward newly appearing targets. The most relevant saccadic metrics are latency, gain and peak velocity.

Anti-saccade task, in which subjects are instructed to look in the direction opposite to a suddenly appearing visual target, provides a measure of inhibitory control and voluntary saccade generation. The most commonly used metric is the error-rate, representing the proportion of trials in which participants fail to suppress the reflexive saccade toward the target.

Fixation task, in which subjects fix their gaze on one or several stationary targets. Common metrics of interest include the rate of saccadic intrusions and the bivariate contour ellipse area (BCEA), which is the area encompassing a given percentage of gaze points.

Smooth pursuit task, in which subjects are instructed to follow a moving target with their gaze. Key metrics include pursuit gain and the number of catch-up saccades.

Impairment in one or more tasks is indicative of neurodegenerative changes or lesions that are disrupting the neural circuits or structures governing those tasks. Detecting these abnormalities through testing may reveal subtle subclinical worsening of physical and/or cognitive disability, which may be used for early detection and diagnosis of neurological diseases, and for evaluating treatment response and disease progression.

Use of eye tracking in neurological diseases

Multiple sclerosis

The focal inflammatory lesions, demyelination and diffuse CNS inflammation that characterize MS are commonly associated with saccadic disorders (dysmetria, adduction slowing in INO), impaired fixation (e.g. gaze-evoked nystagmus, saccadic intrusions and oscillations), impaired smooth pursuit and horizonal gaze paresis [32]. The most common saccadic disorder is INO resulting from demyelination of the MLF. Bilateral INO may result from axonal MLF damage, which impairs smooth-pursuit signalling from the vestibular nuclei to the midbrain nuclei [47]. In addition, cerebellar lesions may be associated with saccadic dysmetria, impaired smooth pursuit and gaze-evoked nystagmus [32]. Smooth-pursuit gain ≤ 70% (normal value ≥ 90%) is suggestive of impaired cerebellar function.

The dorsolateral prefrontal cortex and, to a lesser degree, the cerebellum, are required to inhibit the

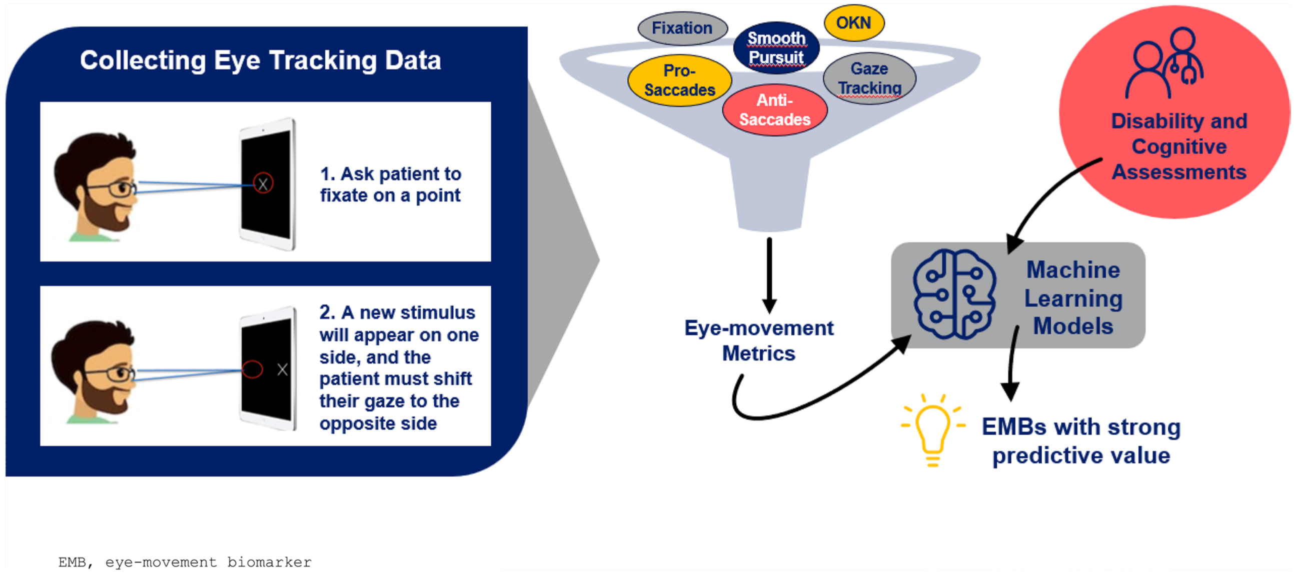

Fig. 2 Workflow for deriving eye-movement biomarkers (EMB) from eye-tracking assessments. During tablet- or camera-based eye-tracking tasks, patients fixate on a central point and shift their gaze in response to appearing targets. Data from multiple paradigms (e.g. pro-saccades, anti-saccades, smooth pursuit, fixation, and gaze-tracking tasks) are processed to extract eye-movement metrics such as saccadic latency, amplitude, velocity, error rate and corrective saccades, fixation stability/microsaccades, and pursuit gain. These metrics can then be analysed using statistical and machine-learning models and related to disability and cognitive outcomes to identify EMB with potential predictive value

automatic pro-saccade response, which makes the anti-saccade task useful to identify cognitive deficits. Anti-saccade errors may be present in radiologically isolated syndrome (RIS) and clinically isolated syndrome (CIS) [47, 48], indicating that eye-movement biomarkers have the potential for early detection of cognitive impairment. Pro-saccade errors and prolonged anti-saccade latencies become increasingly common with MS progression, with error rates significantly correlated with Paced Auditory Serial Addition Test (PASAT) scores [29]. During a two-year follow-up, there was worsening on the anti-saccade task (error rate, latency, accuracy), indicating disease progression that was not apparent with conventional evaluations such as the Expanded Disability Status Scale (EDSS) [49]. Poor performance on the memory-guided saccade test has been shown to be correlated with cognitive testing on the PASAT, Symbol Digit Modalities Test (SDMT) and the California Verbal Learning Test (CVLT), indicating impairments in working memory [50]. Saccadic delay is increasingly prolonged with aging and disease duration [31]. Saccadic latency > 300 ms (normal range 200–250 ms) indicates slowed neural conduction. Errors on more demanding tasks, such as double-step saccadic testing (two targets in succession), have been shown to be associated with more severe whole-brain and grey-matter atrophy [44], suggesting that eye-movement biomarkers may be a useful adjunct to imaging in identifying neurodegeneration.

Abnormalities of fixational stability may provide insights on physical disability worsening. Fixation drift ≥ 0.05 degrees (normal value ≤ 0.01) is associated with attention deficits or motor instability. A higher number of microsaccades (involuntary adjustments to fixed gaze) has been shown to be significantly associated with higher EDSS scores and Functional System Scores for brainstem, cerebellar and pyramidal function [51]. The number, amplitude and peak acceleration of microsaccades are also significantly associated with MS fatigue. Impaired visual fixation reportedly occurs in 24% of MS patients; prevalence was higher with increasing disability [52]. Higher-amplitude square-wave jerks (a form of saccadic intrusion) are associated with poorer vision-related quality of life.

The emergence of novel methods of eye-movement recording employing machine-learning algorithms has permitted a much more sophisticated analysis of multiple EMB and their correlation with physical and cognitive disability. One study reported that nine eye-movement parameters were correlated with EDSS score, nine were correlated with SDMT scores, five were correlated with the Brief International Cognitive Assessment for MS (BICAMS), and 10 were correlated with the MS Functional Composite (MSFC), which comprises the PASAT and two measures of upper limb (9-Hole Peg Test) and lower limb (25-Foot Walk) function [53]. Combining multiple eye-movement parameters explained between 53% (MSFC) and 74% (EDSS) of the variance of the clinical outcome measures. In this study, an algorithm using multiple eye movement metrics correlated better than any individual parameter; the whole was indeed greater than the sum of the parts. These results indicated that serial assessment of eye-movement abnormalities, combined with machine learning to generate composite metrics, may enable clinicians to obtain real-time tracking of physical and cognitive changes in MS in a simple and non-invasive manner.

Optokinetic nystagmus (OKN) responses may also be informative in neurological disease. Abnormal or dissociated OKN has been reported in cortical lesions and in MS, particularly in association with INO and fascicular abducens pathway involvement. Although OKN is less commonly included in eye-tracking protocols, it represents a complementary oculomotor measure that may warrant further exploration in future studies.

Limitations and open questions

In MS, interpretation of EMB findings can be influenced by variability in task paradigms (e.g. pro-/anti-saccade vs. memory-guided tasks) and analytic definitions of key metrics. Disease heterogeneity is another important factor, as disability level, lesion burden, fatigue, visual pathway involvement and history of optic neuritis are not always accounted for and may affect performance independently of cognitive status. Many studies are based on modest or single-centre samples and differ in inclusion criteria and clinical endpoints, which can limit generalizability across MS phenotypes and settings. Future work would benefit from harmonized task protocols, clearer reporting of visual and neurological covariates, and broader validation across cohorts.

Parkinson's disease

A range of eye-movement abnormalities occur throughout the clinical course of Parkinson's disease (PD) as the underlying pathobiology evolves. Hypometria of voluntary saccades (vertical > horizontal) is a common early finding [54]. Impairments in memory-guided saccades, a result of disruption of the frontal cortex-basal ganglia circuit, worsen with ongoing neurodegeneration of the dopaminergic system [55]. Saccade hypometria may be so pronounced that a multiple-step pattern (MSP) is required, which has prompted the suggestion that MSP could be used as a diagnostic biomarker in PD [15]. In the ICICLE-PD study, pro-saccade metrics (latency, duration, amplitude, peak and average velocity) were predictive of cognitive decline in the domains of global cognition, executive function, attention, and memory at 54 months [56]. These results were supported by a study of video-based eye tracking, which reported that eye biomarkers (saccade, pupil, blink) were predictive of cognitive and motor scores [57].

Another common finding is a significantly higher error rate and prolonged reaction times in the anti-saccade task, which evaluates inhibitory control of reflexive saccades mediated by the frontal-striatal network [17]. Increased anti-saccade latency was associated with greater motor dysfunction on the Unified Parkinson's Disease Rating Scale (UPDRS). This suggests that eye-movement abnormalities may serve as a useful alternative to conventional motor assessments as biomarkers of disease progression [58, 59].

Fixation instability, characterized by a higher frequency of square-wave jerks and microsaccades, has also been proposed as a diagnostic biomarker in suspected PD [60]. Fixation requires the interaction of the frontal eye fields that control voluntary eye movements, the basal ganglia to modulate fixation and suppress saccades, and the superior colliculus, all of which require an intact dopaminergic system to maintain their connectivity [61]. Moreover, dopamine depletion impairs smooth pursuit, which is controlled by the parieto-occipital cortex, the frontal eye fields, the basal ganglia and the cerebellum [62].

Vergence eye movements may also be affected in parkinsonian disorders, with impaired convergence and reduced vergence amplitude reported in some patients. Although vergence is less frequently captured in standard eye-tracking tasks, these abnormalities may contribute to near-vision complaints and represent an additional domain of oculomotor dysfunction.

Eye-tracking technologies have shown significant potential in PD screening and diagnosis. One model using four key metrics (fixation accuracy, pro-saccade velocity, anti-saccade accuracy and visual search duration) was shown to be satisfactory for preliminary screening and suitable for clinical practice [62]. A separate study reported that 16 eye-movement parameters were correlated with UPDRS motor scores [63].

Multiple eye-movement biomarkers are also correlated with different domains of cognitive functioning as assessed by the Montreal Cognitive Assessment (MoCA, global cognition), the Trail-Making Test (TMT, attention and working memory), the Controlled Oral Word Association Test of verbal fluency (executive function) and the Hopkins Verbal Learning Test (memory) in PD. These preliminary results demonstrate the significant potential of eye-tracking technologies in the early detection of PD and in the ongoing evaluation of motor and cognitive changes throughout the course of disease. Eye-tracking metrics may also prove useful in clinical trials for evaluating response to treatment or neurorehabilitation strategies [58, 64].

Limitations and open questions

Across PD studies, EMB results are occasionally difficult to compare because eye-tracking tasks, stimulus timing, and device characteristics (e.g. sampling rates, calibration methods) are not consistently standardized, and analysis pipelines differ across groups. Medication state is a major source of variability: dopaminergic ON/OFF status, motor fluctuations, and sedating co-medications are not always reported. Performance may also be influenced by fatigue, depression, and cognitive slowing. In addition, many studies rely on relatively small or single-centre samples, which limits generalizability across populations and clinical settings. Future work should prioritize multisite cohorts with harmonized task design and reporting standards, and more systematic control of medication and state effects.

Alzheimer's disease

Eye-movement abnormalities are common in Alzheimer's disease (AD) but may lack specificity to AD pathology. The most frequent saccade abnormalities are prolonged latency of voluntary saccades, uncorrected errors of anti-saccades and fixation instability. A longer anti-saccade latency and a higher frequency of anti-saccade uncorrected errors differentiate AD from healthy controls, as well as amnesic mild cognitive impairment (MCI) from non-amnesic MCI [7]. Anti-saccade error was the most accurate ocular biomarker for diagnosing MCI and AD in a recent meta-analysis [65]. While the underlying mechanisms are unclear, saccade abnormalities appear to be associated with impairments in attention, working memory and decision-making [4, 66, 67]. Saccadic intrusions, which indicate impaired inhibition of reflexive saccades, have been reported to have greater frequency or amplitude in AD compared to healthy controls [68]. Pro-saccade and anti-saccade metrics are correlated with performance on cognitive tests, including the MoCA and the Mini-Mental State Examination (MMSE) [69]. Pro-saccade latency differentiates AD from healthy controls [4], but anti-saccade tasks appear to be more useful in identifying MCI [70].

During smooth-pursuit tasks, AD patients have lower average peak velocity and a higher number and amplitude of anticipatory saccades [71, 72]. A potential confound is clinical depression, with one study reporting that slower pursuit velocity and impaired fixation were correlated with Hamilton Depression Rating Scale (HAM-17) scores [73]. Lateral fixation accuracy is also impaired in MCI/AD and is correlated with poorer scores on the SDMT and TMT [66, 74].

The use of multiple eye-movement biomarkers has been shown to be highly predictive of AD. One study reported that a combination of the anti-saccade correction rate, anti-saccade velocity, saccade duration, memory saccade accuracy, number of inhibition failures and lateral fixation accuracy had 76.9% accuracy in predicting progression to AD [66]. Other groups have used eye-movement tracking during the performance of visual memory tasks to detect cognitive decline in preclinical and clinical AD [75–77]. Diagnostic and prognostic models are now employing machine learning with more sophisticated algorithms, deep learning and other enhancements to aid in the early detection of cognitive decline [78–80].

Other dementias

The eye-movement abnormalities documented in other dementias largely depend on the cortical or subcortical regions that have been implicated in the generation of saccades. Dementia with Lewy bodies (DLB) is associated with impairments in voluntary pro-saccades/anti-saccades similar to what is seen in AD and PD dementia [11]. What differentiates DLB from AD is impairment of reflexive saccades, which appears to be due to subcortical neurodegeneration [11]. The anti-saccade error rate is similar in DLB and AD, but the correction rate is significantly lower in DLB. Lower saccade accuracy and peak velocity have also been reported in DLB [13].

In frontotemporal lobar degeneration (FTLD), eye-movement abnormalities differ according to the phenotype. Impairments on the anti-saccade task occur in frontotemporal dementia (FTD) and progressive non-fluent aphasia but not in semantic dementia [81]. FTLD patients can self-correct anti-saccade errors, in contrast to AD patients [24]. Early saccades are correlated with atrophy of the left frontal eye field [81]. A slowing of reflexive visually-guided saccades is suggestive of tau pathology in FTD, which may help to differentiate it from non-tau or TDP-43 FTP [8].

The tauopathy progressive supranuclear palsy (PSP) is associated with several unique alterations in vertical saccade velocity and saccade gain that are diagnostic; these measures can be used to distinguish PSP from PD [82–84]. Vertical gaze palsy is typically accompanied by relative preservation of the vestibulo-ocular reflex (VOR), reflecting the supranuclear nature of the deficit. Although high-frequency vestibular responses are not assessed by current tablet- or camera-based eye-tracking platforms, incorporation of simple low-frequency VOR testing could help demonstrate this dissociation in digital oculomotor assessments. A calculation that multiplies saccade amplitude with vertical gaze peak velocity had 91.5% accuracy (specificity 94.7%, sensitivity 84.3%) in differentiating PSP from PD [84]. Also noteworthy are the increased anti-saccade error rates, and worsening memory-guided saccade error rates with disease progression [85]. Eye-movement abnormalities are also correlated with the severity of midbrain atrophy [82].

The subtle nuances or 'EMB signatures' of eye movement abnormalities differ across the dementias and thus combinations of eye movement biomarkers may also have utility in identifying different pathophysiologies associated with the clinical presentations, aiding in diagnostic precision.

Limitations and open questions

In AD and related dementias, as in the other neurological conditions covered in this review, EMB results can be shaped by differences in task design and stimulus complexity. Variability in how cognitive status and disease stage are defined across studies can also affect the interpretation and comparability of findings. Performance on oculomotor tasks may be influenced by age, visual acuity, depression, apathy, and psychotropic medications, which are not always measured or controlled in existing cohorts. Diagnostic groups are sometimes small or clinically heterogeneous, and cross-sectional designs predominate, making it difficult to determine specificity of findings across dementia subtypes. Larger, phenotypically well-characterized cohorts and comparative studies across syndromes will be important to clarify the diagnostic and prognostic value of EMB.

Amyotrophic lateral sclerosis

The eye-movement abnormalities that occur in amyotrophic lateral sclerosis (ALS) depend on the location and severity of the underlying neuropathology. Common deficits in ALS patients include abnormalities in saccades and smooth pursuit; frequency and severity are correlated with the degree of cognitive impairment, suggesting an association with frontal oculomotor regions [86]. Square-wave jerks and abnormal cogwheeling (jerking motion) during smooth pursuit are especially common in bulbar-onset ALS [87]. Increases in the anti-saccade error rate and prolonged latencies are associated with executive dysfunction and indicative of reduced dorsolateral prefrontal cortex activity [86, 88]. There are also prolonged latencies and increased errors in memory-guided saccades resulting from impairments in the prefrontal cortex as well as the frontal and parietal eye fields [89].

Saccade abnormalities differ in bulbar-onset ALS compared to spinal-onset ALS. Bulbar-onset ALS is characterized by slow visually-guided saccades and smooth-pursuit deficits, which may indicate involvement of cortical-brainstem circuits, whereas spinal-onset ALS demonstrates short saccades indicative of dysfunction in the parietal-superior colliculus-cerebellar network [90, 91]. As eye-movement abnormalities become increasingly impaired with progressive involvement of the prefrontal, midbrain and brainstem regions, EMB may be used as a biomarker of neurodegenerative changes and disease progression in advanced ALS [91]. A clinical challenge, however, is that impaired oculomotor function may impede the use of communication devices that use eye tracking in late-stage ALS [92].

Implementation in practice

Early eye tracking in the 1950 s employed a magnetic search coil (SC) technique, in which a coil of wire embedded in a contact lens interacted with an external magnetic field to ascertain eye position and movement at a sampling rate of 1000 Hz. Electro-oculography (EOG) used electrodes positioned around the eye to measure changes in the electrical field caused by eye movements. These methods required expensive equipment that was not widely accessible and were invasive and unpleasant for patients. Less invasive infra-red oculography (IOG) was subsequently shown to be as accurate as the SC method and offered a range of sampling rates (250–1000 Hz) [93].

In recent years, video oculography (VOG) systems have emerged that use cameras and proprietary machine-learning algorithms to analyse eye movements without the need for measuring head movements. The commercial availability of low-cost digital cameras in portable devices has enabled these systems to be used more widely in a variety of settings to improve patient access and convenience. Eye tracking may be embedded in goggles or head-mounted displays (HMD) (sampling rate 120 Hz) [94], smart phones [95], and iPad tablets [53, 63]. HMD displays need to be updated frequently, which is dependent on how rapidly the screen detects a saccade (based on sampling frequency and the algorithm used) and the time needed to update the screen (based on rendering time, frame rate and the video hardware) [95]. A comparison of commercially-available HMD systems found significant differences in tracking delays (from 15–50 ms) and end-to-end latencies (from 45–80 ms) [96]. Smart phone eye recordings reported limited accuracy of eye recordings (23% error); accuracy was somewhat better for horizontal compared to vertical movements (17% vs. 27% error). While this may indicate that some systems are less appropriate for medical applications, ongoing improvements can address these technological limitations. The lower accuracy of smart phone systems may be off-set by their convenience to patients and wider accessibility in clinical practice. The acceptable trade-off of accuracy and accessibility has not yet been determined.

Two clinical studies to date have employed a tablet-based system that uses a commercially-available iPad Pro with proprietary software installed [53, 63]. Eye movements are recorded with the tablet camera at 60 frames/sec. Accuracy of the gaze-tracking algorithms is 0.47 degrees (difference between the actual and recorded gaze positions), and precision of the sample points is 0.33 degrees, which is similar to that of IOG systems. The software detects the user's facial landmarks and eye position, which eliminates the need for restraints (e.g. chin guard) and individualizes the gaze model. A study is ongoing to evaluate eye tracking as a biomarker of disease progression in MS and the feasibility of incorporating the technology into clinical practice (NCT05277740). Such systems may provide clinicians with an accessible, cost-effective technology that can be used in routine assessments.

Although EMB technologies show promise as accessible digital assessment tools, evidence from formal cost-effectiveness analyses, regulatory validation, and real-world implementation studies remains limited. Accordingly, their use in clinical practice should be considered exploratory and complementary to existing standards of care.

Future implications for clinical practice

In recent years, considerable advances have been made in our understanding of the underlying pathophysiology of neurodegenerative diseases such as MS, PD, AD and ALS. The use of fluid biomarkers, such as neurofilament-light chain (NfL) and glial fibrillary acidic protein (GFAP), and advanced imaging techniques, such as susceptibility-weighted imaging (SWI) and positron emission tomography (PET), have enabled the earlier detection and improved characterization of the neurodegenerative processes that contribute to disease progression. For example, elevated levels of NfL, a marker of axonal damage, have been detected six years prior to the onset of clinical symptoms in MS [97]. Similarly, amyloid biomarkers (e.g. amyloid PET) are validated methods of identifying early amyloid-beta deposition in the brain in AD [98].

Eye-movement biomarkers have the potential to play an equally important role in the early identification of demyelination and neurodegeneration affecting the cortex, basal ganglia, superior colliculus and cerebellum across a range of neurodegenerative conditions (Fig. 2). For example, anti-saccade errors may be detected before the clinical onset of MS [47, 48], while worsening performance on anti-saccade tasks has been shown to be a more sensitive measure of disease progression than clinical evaluations [49]. During the clinical course, eye-movement biomarkers can differentiate different patterns of neurodegeneration and detect gradual worsening of cognitive and motor function. Ongoing trials are evaluating the utility of this technology across a range of neurological and neuropsychiatric conditions, such as traumatic brain injury, autism spectrum disorder, schizophrenia and mood disorders [99–102].

With advances in the evaluation of neurodegenerative diseases, it is increasingly incumbent upon neurologists to conduct more rigorous and timely assessments. Current tools, such as the EDSS and the UPDRS, are largely subjective measures with high intra- and inter-rater variability that lack the precision to objectively identify neurological deficits. Frequent tracking of disease progression is hindered by time constraints for physicians and patients, and the costs associated with laboratory and radiological testing. The time burden of clinic visits, laboratory testing and imaging may be considerable for patients.

Eye-movement biomarkers present a potential solution to this challenge. Assessments fulfill the clinician's need for additional accurate, sensitive tools to aid in disease surveillance, prognosis and to evaluate the extent and severity of disease. Testing can be performed by allied health professionals, such as pharmacists, nurses or trained medical technicians, in different settings, such as private clinics, community health centres or pharmacies. With minimal instruction, patients and caregivers could conduct eye-movement testing in their own home. Test results could then be analysed by AI-powered algorithms and forwarded electronically to the consulting neurologist for clinical interpretation. This ability to engage patients in a community setting or at home would streamline disease monitoring and relieve pressure on overextended specialty services. Thus, the cost effectiveness, non-invasive nature, and accessibility of this technology would permit more frequent and rapid evaluations that could serve as a vital adjunct to more conventional testing to aid physicians in their clinical decision-making. The equipment required, such as standard smart phones or tablets, is widely available and inexpensive to obtain or maintain, allowing this technology to be scalable in all health care systems as well.

Several studies in this field, including some cited here, rely on proprietary eye-tracking hardware and software pipelines. As a result, EMB performance and derived metrics may not generalize directly across platforms or implementations. Independent validation in external cohorts, cross-platform comparisons with alternative eye-tracking technologies, and greater transparency in task design and analytic methods will be essential to establish reproducibility, interoperability, and clinical robustness. The studies reviewed should therefore be interpreted as evidence of feasibility and emerging utility rather than endorsement of any specific commercial system.

Future implications for clinical research

Just as EMB biomarkers offer a sensitive and potentially cost-effective approach for detecting subtle neurological and cognitive changes in clinical practice, they can also be applied in clinical research settings as potential outcome measures in early-phase clinical trials. By quantifying and combining small alterations in eye movements and patterns, attention and visual processing, these digital biomarkers can reveal early or subclinical disease signals in a rapid and non-invasive manner. Moreover, as inexpensive and non-invasive methods of clinical assessment, they could be deployed more frequently in diverse clinical settings, accelerating the evaluation of candidate molecules in neurodegenerative diseases. Their sensitivity and scalability may improve trial efficiency, reduce sample size requirements, support earlier go/no-go decisions, and ultimately lower development costs.

Conclusion

The past decade has seen a rapid evolution in the systems used to record eye movements, and in the increasingly sophisticated software and machine-learning algorithms that are used to process and analyse the data. Innovations in eye-tracking technology can achieve the goals of greater accuracy and precision in disease monitoring while enabling patient-level access to improved care. Recent clinical trials have also demonstrated that eye-movement biomarkers are uniquely sensitive in detecting clinical progression [103], which may improve the efficiency of drug trials, reduce sample size requirements and lower the cost of clinical testing.

The current revolution in digital biomarkers holds the promise of providing more timely, accurate and accessible assessments of the clinical course of illness across the spectrum of neurodegenerative diseases.

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Declarations

Conflicts of interest Paul S. Giacomini is an employee of Innodem Neurosciences. Patrice Voss is an employee of Innodem Neurosciences. Virginia Devonshire has no competing interests. Raphael Schneider has no competing interests. Gabrielle Macaron has no competing interests. Shamiza Hussein is an employee of Novartis Pharmaceuticals Canada. François Blanchette is an employee of Novartis Pharmaceuticals Canada. Étienne de Villers-Sidani is an employee of Innodem Neurosciences.

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